

Supplementary Information

An Accuracy and Calibration Benchmark of scGPT Pipelines and Matched Classical Models for Cell Type Annotation

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Contents

This file reports the complete outer-fold results, pooled out-of-fold metrics, paired donor-cluster bootstrap comparisons, hyperparameter selections, calibration sensitivity, strict label-scarcity analysis, per-class error analysis, and the separately designated pig-pancreas cross-species stress test. The machine-readable source tables and prediction-level audit files are included in the accompanying reproducibility archive.

1 Completed human benchmark

Table S1: Completed out-of-fold design audit.

Dataset	Donor IDs	Classes	OOF cells	Matched HVGs
TNBC	101	7	427,813	1167–1172
PBMC	199	13	462,034	1132–1141
Brain	12	10	75,580	930–953

Outer-fold variation across datasets and classifiers

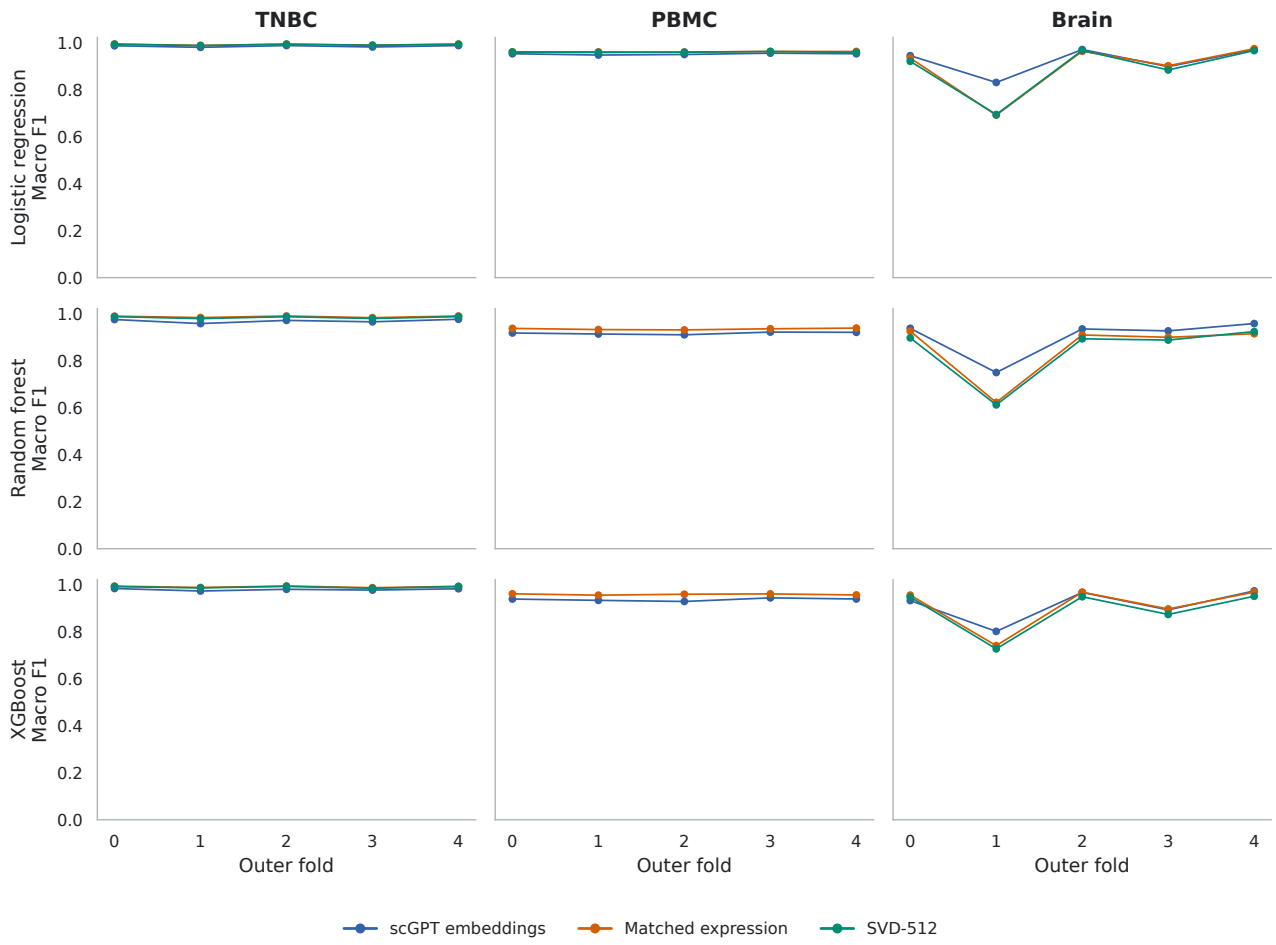


Figure S1: Outer-fold Macro F1 for every human representation–classifier pipeline. Training folds overlap, but each outer-test donor and cell occurs in exactly one fold.

2 Five-fold pipeline coverage

Table S2: Pipeline coverage used by the local aggregation. A pipeline was included only when all five aligned donor-held-out prediction folds were durably available.

Dataset	Representation	Classifier	Status
Brain	Matched expression	Logistic regression	Included
Brain	Matched expression	Random forest	Included
Brain	Matched expression	XGBoost	Included
Brain	scGPT embeddings	Logistic regression	Included
Brain	scGPT embeddings	Random forest	Included
Brain	scGPT embeddings	XGBoost	Included
Brain	SVD-512	Logistic regression	Included
Brain	SVD-512	Random forest	Included
Brain	SVD-512	XGBoost	Included
PBMC	Matched expression	Logistic regression	Included
PBMC	Matched expression	Random forest	Included
PBMC	Matched expression	XGBoost	Included
PBMC	scGPT embeddings	Logistic regression	Included
PBMC	scGPT embeddings	Random forest	Included
PBMC	scGPT embeddings	XGBoost	Included
PBMC	SVD-512	Logistic regression	Included
PBMC	SVD-512	Random forest	Not available
PBMC	SVD-512	XGBoost	Not available
TNBC	Matched expression	Logistic regression	Included
TNBC	Matched expression	Random forest	Included
TNBC	Matched expression	XGBoost	Included
TNBC	scGPT embeddings	Logistic regression	Included
TNBC	scGPT embeddings	Random forest	Included
TNBC	scGPT embeddings	XGBoost	Included
TNBC	SVD-512	Logistic regression	Included
TNBC	SVD-512	Random forest	Included
TNBC	SVD-512	XGBoost	Included

3 Pooled out-of-fold metrics

Table S3: Complete pooled out-of-fold metrics for the three human atlases. SVD-512 cells are blank in the main table where a five-fold classifier pipeline was not available.

Dataset	Representation	Classifier	n	Macro F1	Accuracy	Balanced acc.	ECE	Brier	NLL
Brain	Matched expression	Logistic regression	75580	0.911	0.985	0.874	0.004	0.027	0.107
Brain	scGPT embeddings	Logistic regression	75580	0.940	0.986	0.915	0.003	0.025	0.070
Brain	SVD-512	Logistic regression	75580	0.906	0.985	0.872	0.004	0.027	0.106
Brain	Matched expression	Random forest	75580	0.870	0.983	0.807	0.021	0.032	0.156
Brain	scGPT embeddings	Random forest	75580	0.908	0.984	0.859	0.020	0.032	0.132
Brain	SVD-512	Random forest	75580	0.848	0.985	0.790	0.011	0.028	0.160
Brain	Matched expression	XGBoost	75580	0.925	0.987	0.887	0.008	0.024	0.072
Brain	scGPT embeddings	XGBoost	75580	0.930	0.986	0.903	0.009	0.026	0.081
Brain	SVD-512	XGBoost	75580	0.904	0.987	0.856	0.010	0.024	0.092
PBMC	Matched expression	Logistic regression	462034	0.959	0.953	0.957	0.004	0.069	0.121
PBMC	scGPT embeddings	Logistic regression	462034	0.949	0.940	0.947	0.002	0.089	0.159
PBMC	SVD-512	Logistic regression	462034	0.958	0.952	0.955	0.003	0.071	0.125
PBMC	Matched expression	Random forest	462034	0.933	0.941	0.914	0.039	0.094	0.186
PBMC	scGPT embeddings	Random forest	462034	0.915	0.907	0.897	0.059	0.148	0.281
PBMC	Matched expression	XGBoost	462034	0.957	0.953	0.953	0.003	0.070	0.123
PBMC	scGPT embeddings	XGBoost	462034	0.935	0.925	0.929	0.004	0.111	0.199
TNBC	Matched expression	Logistic regression	427813	0.990	0.988	0.988	0.000	0.019	0.038
TNBC	scGPT embeddings	Logistic regression	427813	0.983	0.979	0.981	0.001	0.033	0.065
TNBC	SVD-512	Logistic regression	427813	0.989	0.987	0.988	0.001	0.020	0.040
TNBC	Matched expression	Random forest	427813	0.985	0.984	0.978	0.037	0.031	0.078
TNBC	scGPT embeddings	Random forest	427813	0.968	0.964	0.953	0.038	0.061	0.134
TNBC	SVD-512	Random forest	427813	0.982	0.979	0.977	0.023	0.037	0.081
TNBC	Matched expression	XGBoost	427813	0.990	0.989	0.987	0.001	0.018	0.035
TNBC	scGPT embeddings	XGBoost	427813	0.978	0.974	0.972	0.001	0.040	0.077
TNBC	SVD-512	XGBoost	427813	0.988	0.986	0.986	0.003	0.022	0.043

4 Outer-fold metrics

Table S4: Outer-fold metrics. Fold-level values are not independent because their training partitions overlap.

Dataset	Fold	Representation	Classifier	Groups	Cells	Macro F1	Accuracy	ECE	Brier
Brain	0	Matched expression	Logistic regression	3	19314	0.932	0.973	0.024	0.051
Brain	1	Matched expression	Logistic regression	1	11034	0.690	0.966	0.008	0.053
Brain	2	Matched expression	Logistic regression	3	15401	0.961	0.994	0.005	0.011
Brain	3	Matched expression	Logistic regression	3	14244	0.900	0.994	0.006	0.012
Brain	4	Matched expression	Logistic regression	2	15587	0.972	0.996	0.013	0.008
Brain	0	scGPT embeddings	Logistic regression	3	19314	0.943	0.978	0.015	0.043
Brain	1	scGPT embeddings	Logistic regression	1	11034	0.829	0.973	0.012	0.045
Brain	2	scGPT embeddings	Logistic regression	3	15401	0.969	0.994	0.008	0.010
Brain	3	scGPT embeddings	Logistic regression	3	14244	0.896	0.992	0.012	0.016
Brain	4	scGPT embeddings	Logistic regression	2	15587	0.968	0.994	0.006	0.010
Brain	0	SVD-512	Logistic regression	3	19314	0.919	0.973	0.023	0.051
Brain	1	SVD-512	Logistic regression	1	11034	0.693	0.966	0.008	0.052
Brain	2	SVD-512	Logistic regression	3	15401	0.965	0.994	0.005	0.010
Brain	3	SVD-512	Logistic regression	3	14244	0.882	0.993	0.001	0.012
Brain	4	SVD-512	Logistic regression	2	15587	0.964	0.995	0.018	0.009
Brain	0	Matched expression	Random forest	3	19314	0.924	0.977	0.013	0.045
Brain	1	Matched expression	Random forest	1	11034	0.621	0.960	0.021	0.067
Brain	2	Matched expression	Random forest	3	15401	0.908	0.990	0.027	0.018
Brain	3	Matched expression	Random forest	3	14244	0.898	0.991	0.023	0.020
Brain	4	Matched expression	Random forest	2	15587	0.913	0.994	0.033	0.016
Brain	0	scGPT embeddings	Random forest	3	19314	0.936	0.977	0.016	0.046
Brain	1	scGPT embeddings	Random forest	1	11034	0.748	0.966	0.008	0.057
Brain	2	scGPT embeddings	Random forest	3	15401	0.933	0.990	0.036	0.021
Brain	3	scGPT embeddings	Random forest	3	14244	0.925	0.992	0.030	0.021
Brain	4	scGPT embeddings	Random forest	2	15587	0.956	0.993	0.025	0.016
Brain	0	SVD-512	Random forest	3	19314	0.895	0.978	0.018	0.045
Brain	1	SVD-512	Random forest	1	11034	0.611	0.964	0.003	0.055
Brain	2	SVD-512	Random forest	3	15401	0.891	0.993	0.020	0.015
Brain	3	SVD-512	Random forest	3	14244	0.886	0.993	0.017	0.015
Brain	4	SVD-512	Random forest	2	15587	0.922	0.995	0.021	0.013
Brain	0	Matched expression	XGBoost	3	19314	0.954	0.979	0.019	0.041
Brain	1	Matched expression	XGBoost	1	11034	0.739	0.970	0.020	0.051
Brain	2	Matched expression	XGBoost	3	15401	0.966	0.995	0.004	0.009
Brain	3	Matched expression	XGBoost	3	14244	0.895	0.992	0.003	0.014
Brain	4	Matched expression	XGBoost	2	15587	0.967	0.995	0.001	0.008
Brain	0	scGPT embeddings	XGBoost	3	19314	0.931	0.977	0.019	0.043
Brain	1	scGPT embeddings	XGBoost	1	11034	0.800	0.971	0.025	0.053
Brain	2	scGPT embeddings	XGBoost	3	15401	0.965	0.994	0.003	0.011
Brain	3	scGPT embeddings	XGBoost	3	14244	0.891	0.992	0.002	0.015
Brain	4	scGPT embeddings	XGBoost	2	15587	0.972	0.994	0.003	0.009
Brain	0	SVD-512	XGBoost	3	19314	0.946	0.978	0.020	0.042
Brain	1	SVD-512	XGBoost	1	11034	0.726	0.971	0.024	0.053
Brain	2	SVD-512	XGBoost	3	15401	0.947	0.994	0.003	0.010
Brain	3	SVD-512	XGBoost	3	14244	0.872	0.993	0.005	0.012
Brain	4	SVD-512	XGBoost	2	15587	0.949	0.995	0.002	0.008
PBMC	0	Matched expression	Logistic regression	40	93930	0.959	0.953	0.005	0.070
PBMC	1	Matched expression	Logistic regression	38	92275	0.959	0.952	0.002	0.071
PBMC	2	Matched expression	Logistic regression	42	93316	0.958	0.952	0.004	0.071
PBMC	3	Matched expression	Logistic regression	38	90015	0.960	0.955	0.004	0.066
PBMC	4	Matched expression	Logistic regression	41	92498	0.960	0.955	0.004	0.066
PBMC	0	scGPT embeddings	Logistic regression	40	93930	0.951	0.940	0.003	0.090
PBMC	1	scGPT embeddings	Logistic regression	38	92275	0.945	0.937	0.003	0.093
PBMC	2	scGPT embeddings	Logistic regression	42	93316	0.948	0.938	0.002	0.093
PBMC	3	scGPT embeddings	Logistic regression	38	90015	0.953	0.944	0.001	0.085
PBMC	4	scGPT embeddings	Logistic regression	41	92498	0.951	0.943	0.002	0.085
PBMC	0	SVD-512	Logistic regression	40	93930	0.958	0.951	0.003	0.072
PBMC	1	SVD-512	Logistic regression	38	92275	0.956	0.949	0.002	0.074
PBMC	2	SVD-512	Logistic regression	42	93316	0.957	0.951	0.002	0.072
PBMC	3	SVD-512	Logistic regression	38	90015	0.960	0.955	0.002	0.068
PBMC	4	SVD-512	Logistic regression	41	92498	0.956	0.953	0.006	0.069
PBMC	0	Matched expression	Random forest	40	93930	0.935	0.941	0.037	0.094
PBMC	1	Matched expression	Random forest	38	92275	0.931	0.937	0.040	0.099
PBMC	2	Matched expression	Random forest	42	93316	0.929	0.939	0.041	0.097
PBMC	3	Matched expression	Random forest	38	90015	0.934	0.944	0.039	0.090
PBMC	4	Matched expression	Random forest	41	92498	0.937	0.944	0.037	0.089
PBMC	0	scGPT embeddings	Random forest	40	93930	0.916	0.907	0.059	0.149
PBMC	1	scGPT embeddings	Random forest	38	92275	0.912	0.904	0.057	0.151
PBMC	2	scGPT embeddings	Random forest	42	93316	0.908	0.901	0.064	0.159

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Table S4 continued

Dataset	Fold	Representation	Classifier	Groups	Cells	Macro F1	Accuracy	ECE	Brier
PBMC	3	scGPT embeddings	Random forest	38	90015	0.920	0.912	0.056	0.141
PBMC	4	scGPT embeddings	Random forest	41	92498	0.919	0.912	0.057	0.141
PBMC	0	Matched expression	XGBoost	40	93930	0.960	0.953	0.004	0.069
PBMC	1	Matched expression	XGBoost	38	92275	0.954	0.950	0.005	0.073
PBMC	2	Matched expression	XGBoost	42	93316	0.958	0.953	0.002	0.070
PBMC	3	Matched expression	XGBoost	38	90015	0.959	0.956	0.003	0.065
PBMC	4	Matched expression	XGBoost	41	92498	0.955	0.952	0.007	0.071
PBMC	0	scGPT embeddings	XGBoost	40	93930	0.937	0.927	0.008	0.109
PBMC	1	scGPT embeddings	XGBoost	38	92275	0.932	0.923	0.010	0.114
PBMC	2	scGPT embeddings	XGBoost	42	93316	0.927	0.916	0.006	0.124
PBMC	3	scGPT embeddings	XGBoost	38	90015	0.942	0.930	0.003	0.103
PBMC	4	scGPT embeddings	XGBoost	41	92498	0.937	0.930	0.007	0.104
TNBC	0	Matched expression	Logistic regression	19	85988	0.992	0.990	0.001	0.015
TNBC	1	Matched expression	Logistic regression	20	84601	0.986	0.984	0.003	0.025
TNBC	2	Matched expression	Logistic regression	21	86821	0.991	0.990	0.006	0.016
TNBC	3	Matched expression	Logistic regression	21	86079	0.987	0.985	0.002	0.024
TNBC	4	Matched expression	Logistic regression	20	84324	0.991	0.989	0.001	0.017
TNBC	0	scGPT embeddings	Logistic regression	19	85988	0.985	0.982	0.004	0.028
TNBC	1	scGPT embeddings	Logistic regression	20	84601	0.978	0.972	0.003	0.042
TNBC	2	scGPT embeddings	Logistic regression	21	86821	0.985	0.982	0.001	0.028
TNBC	3	scGPT embeddings	Logistic regression	21	86079	0.979	0.976	0.005	0.037
TNBC	4	scGPT embeddings	Logistic regression	20	84324	0.985	0.981	0.001	0.029
TNBC	0	SVD-512	Logistic regression	19	85988	0.991	0.989	0.001	0.016
TNBC	1	SVD-512	Logistic regression	20	84601	0.985	0.983	0.003	0.027
TNBC	2	SVD-512	Logistic regression	21	86821	0.991	0.989	0.001	0.017
TNBC	3	SVD-512	Logistic regression	21	86079	0.987	0.984	0.002	0.025
TNBC	4	SVD-512	Logistic regression	20	84324	0.991	0.989	0.001	0.018
TNBC	0	Matched expression	Random forest	19	85988	0.987	0.986	0.037	0.027
TNBC	1	Matched expression	Random forest	20	84601	0.981	0.982	0.037	0.034
TNBC	2	Matched expression	Random forest	21	86821	0.988	0.988	0.039	0.027
TNBC	3	Matched expression	Random forest	21	86079	0.981	0.979	0.036	0.040
TNBC	4	Matched expression	Random forest	20	84324	0.988	0.987	0.038	0.027
TNBC	0	scGPT embeddings	Random forest	19	85988	0.973	0.970	0.043	0.053
TNBC	1	scGPT embeddings	Random forest	20	84601	0.956	0.957	0.038	0.072
TNBC	2	scGPT embeddings	Random forest	21	86821	0.969	0.964	0.035	0.059
TNBC	3	scGPT embeddings	Random forest	21	86079	0.963	0.959	0.029	0.066
TNBC	4	scGPT embeddings	Random forest	20	84324	0.974	0.971	0.044	0.052
TNBC	0	SVD-512	Random forest	19	85988	0.985	0.982	0.025	0.032
TNBC	1	SVD-512	Random forest	20	84601	0.977	0.973	0.021	0.045
TNBC	2	SVD-512	Random forest	21	86821	0.985	0.982	0.029	0.035
TNBC	3	SVD-512	Random forest	21	86079	0.978	0.974	0.018	0.042
TNBC	4	SVD-512	Random forest	20	84324	0.985	0.983	0.023	0.030
TNBC	0	Matched expression	XGBoost	19	85988	0.992	0.991	0.002	0.014
TNBC	1	Matched expression	XGBoost	20	84601	0.986	0.986	0.001	0.021
TNBC	2	Matched expression	XGBoost	21	86821	0.993	0.992	0.002	0.013
TNBC	3	Matched expression	XGBoost	21	86079	0.985	0.984	0.003	0.024
TNBC	4	Matched expression	XGBoost	20	84324	0.991	0.990	0.005	0.016
TNBC	0	scGPT embeddings	XGBoost	19	85988	0.982	0.980	0.001	0.031
TNBC	1	scGPT embeddings	XGBoost	20	84601	0.971	0.968	0.006	0.048
TNBC	2	scGPT embeddings	XGBoost	21	86821	0.979	0.975	0.004	0.039
TNBC	3	scGPT embeddings	XGBoost	21	86079	0.975	0.971	0.010	0.045
TNBC	4	scGPT embeddings	XGBoost	20	84324	0.981	0.976	0.005	0.036
TNBC	0	SVD-512	XGBoost	19	85988	0.990	0.988	0.005	0.018
TNBC	1	SVD-512	XGBoost	20	84601	0.985	0.982	0.003	0.028
TNBC	2	SVD-512	XGBoost	21	86821	0.991	0.990	0.000	0.016
TNBC	3	SVD-512	XGBoost	21	86079	0.983	0.981	0.002	0.029
TNBC	4	SVD-512	XGBoost	20	84324	0.990	0.988	0.006	0.019

5 SVD-512 dimensionality control

Table S5: Fold-specific SVD-512 audit. Decomposition parameters were fitted without outer-test cells.

Dataset	Fold	Components	Fit cells	Explained variance	Training only
Brain	0	512	56,266	0.935	Yes
Brain	1	512	64,547	0.939	Yes
Brain	2	512	60,179	0.942	Yes
Brain	3	512	61,336	0.939	Yes
Brain	4	512	59,996	0.934	Yes
PBMC	0	512	100,000	0.959	Yes
PBMC	1	512	100,000	0.958	Yes
PBMC	2	512	100,000	0.958	Yes
PBMC	3	512	100,000	0.959	Yes
PBMC	4	512	100,000	0.959	Yes
TNBC	0	512	100,000	0.954	Yes
TNBC	1	512	100,000	0.954	Yes
TNBC	2	512	100,000	0.954	Yes
TNBC	3	512	100,000	0.956	Yes
TNBC	4	512	100,000	0.955	Yes

6 Donor-cluster bootstrap comparisons

Table S6: All paired donor-cluster bootstrap contrasts. Intervals condition on the fitted pipelines.

Dataset	Classifier	Contrast	A	B	$A - B$	95% interval	$P(A > B)$
Brain	Logistic regression	Expression – scGPT	0.911	0.940	-0.028	[-0.072, 0.004]	0.081
Brain	Random forest	Expression – scGPT	0.870	0.908	-0.039	[-0.066, -0.014]	0.000
Brain	XGBoost	Expression – scGPT	0.925	0.930	-0.005	[-0.024, 0.014]	0.334
PBMC	Logistic regression	Expression – scGPT	0.959	0.949	0.010	[0.009, 0.011]	1.000
PBMC	Random forest	Expression – scGPT	0.933	0.915	0.018	[0.016, 0.020]	1.000
PBMC	XGBoost	Expression – scGPT	0.957	0.935	0.022	[0.020, 0.024]	1.000
TNBC	Logistic regression	Expression – scGPT	0.990	0.983	0.007	[0.006, 0.008]	1.000
TNBC	Random forest	Expression – scGPT	0.985	0.968	0.018	[0.014, 0.022]	1.000
TNBC	XGBoost	Expression – scGPT	0.990	0.978	0.012	[0.010, 0.014]	1.000
Brain	Logistic regression	Expression – SVD	0.911	0.906	0.005	[-0.001, 0.012]	0.950
Brain	Random forest	Expression – SVD	0.870	0.848	0.022	[0.009, 0.035]	1.000
Brain	XGBoost	Expression – SVD	0.925	0.904	0.021	[0.013, 0.030]	1.000
PBMC	Logistic regression	Expression – SVD	0.959	0.958	0.002	[0.001, 0.003]	1.000
TNBC	Logistic regression	Expression – SVD	0.990	0.989	0.001	[0.000, 0.001]	1.000
TNBC	Random forest	Expression – SVD	0.985	0.982	0.003	[0.002, 0.005]	1.000
TNBC	XGBoost	Expression – SVD	0.990	0.988	0.002	[0.001, 0.003]	0.998
Brain	Logistic regression	SVD – scGPT	0.906	0.940	-0.033	[-0.075, -0.003]	0.003
Brain	Random forest	SVD – scGPT	0.848	0.908	-0.060	[-0.096, -0.030]	0.000
Brain	XGBoost	SVD – scGPT	0.904	0.930	-0.026	[-0.048, -0.002]	0.016
PBMC	Logistic regression	SVD – scGPT	0.958	0.949	0.008	[0.007, 0.009]	1.000
TNBC	Logistic regression	SVD – scGPT	0.989	0.983	0.007	[0.005, 0.008]	1.000
TNBC	Random forest	SVD – scGPT	0.982	0.968	0.015	[0.012, 0.018]	1.000
TNBC	XGBoost	SVD – scGPT	0.988	0.978	0.010	[0.009, 0.012]	1.000

7 Training-only model selection

Table S7: Frequencies of independently selected parameter sets across the five outer folds.

Dataset	Representation	Classifier	Selected parameter set	Folds
Brain	Matched expression	Logistic regression	{"C": 0.001, "scale": false}	3
Brain	Matched expression	Logistic regression	{"C": 1.0, "scale": false}	1
Brain	Matched expression	Logistic regression	{"C": 0.01, "scale": false}	1
Brain	Matched expression	Random forest	{"max_features": 0.25, "min_samples_leaf": 1}	5
Brain	Matched expression	XGBoost	{"learning_rate": 0.1, "max_depth": 3, "n_estimators": 300}	2
Brain	Matched expression	XGBoost	{"learning_rate": 0.05, "max_depth": 6, "n_estimators": 500}	1
Brain	Matched expression	XGBoost	{"learning_rate": 0.1, "max_depth": 6, "n_estimators": 300}	1
Brain	Matched expression	XGBoost	{"learning_rate": 0.05, "max_depth": 9, "n_estimators": 400}	1
Brain	scGPT embeddings	Logistic regression	{"C": 0.01, "scale": true}	2
Brain	scGPT embeddings	Logistic regression	{"C": 100.0, "scale": false}	1
Brain	scGPT embeddings	Logistic regression	{"C": 10.0, "scale": false}	1
Brain	scGPT embeddings	Logistic regression	{"C": 1000.0, "scale": false}	1
Brain	scGPT embeddings	Random forest	{"max_features": 0.25, "min_samples_leaf": 1}	4
Brain	scGPT embeddings	Random forest	{"max_features": "sqrt", "min_samples_leaf": 5}	1
Brain	scGPT embeddings	XGBoost	{"learning_rate": 0.1, "max_depth": 3, "n_estimators": 300}	2
Brain	scGPT embeddings	XGBoost	{"learning_rate": 0.05, "max_depth": 6, "n_estimators": 500}	2
Brain	scGPT embeddings	XGBoost	{"learning_rate": 0.1, "max_depth": 6, "n_estimators": 300}	1
Brain	SVD-512	Logistic regression	{"C": 0.01, "scale": false}	2
Brain	SVD-512	Logistic regression	{"C": 1.0, "scale": false}	1
Brain	SVD-512	Logistic regression	{"C": 0.001, "scale": false}	1
Brain	SVD-512	Logistic regression	{"C": 0.01, "scale": true}	1
Brain	SVD-512	Random forest	{"max_features": 0.25, "min_samples_leaf": 1}	5
Brain	SVD-512	XGBoost	{"learning_rate": 0.1, "max_depth": 6, "n_estimators": 300}	3
Brain	SVD-512	XGBoost	{"learning_rate": 0.05, "max_depth": 6, "n_estimators": 500}	2
TNBC	Matched expression	Logistic regression	{"C": 0.01, "scale": false}	3
TNBC	Matched expression	Logistic regression	{"C": 0.001, "scale": true}	1
TNBC	Matched expression	Logistic regression	{"C": 0.1, "scale": false}	1
TNBC	Matched expression	Random forest	{"max_features": "sqrt", "min_samples_leaf": 1}	5
TNBC	Matched expression	XGBoost	{"learning_rate": 0.1, "max_depth": 6, "n_estimators": 300}	2
TNBC	Matched expression	XGBoost	{"learning_rate": 0.05, "max_depth": 6, "n_estimators": 500}	1
TNBC	Matched expression	XGBoost	{"learning_rate": 0.05, "max_depth": 9, "n_estimators": 400}	1
TNBC	Matched expression	XGBoost	{"learning_rate": 0.1, "max_depth": 3, "n_estimators": 300}	1
TNBC	scGPT embeddings	Logistic regression	{"C": 0.1, "scale": true}	4
TNBC	scGPT embeddings	Logistic regression	{"C": 0.01, "scale": true}	1
TNBC	scGPT embeddings	Random forest	{"max_features": 0.25, "min_samples_leaf": 1}	5
TNBC	scGPT embeddings	XGBoost	{"learning_rate": 0.1, "max_depth": 6, "n_estimators": 300}	2
TNBC	scGPT embeddings	XGBoost	{"learning_rate": 0.1, "max_depth": 3, "n_estimators": 300}	2
TNBC	scGPT embeddings	XGBoost	{"learning_rate": 0.05, "max_depth": 6, "n_estimators": 500}	1
TNBC	SVD-512	Logistic regression	{"C": 0.01, "scale": false}	4
TNBC	SVD-512	Logistic regression	{"C": 0.1, "scale": false}	1
TNBC	SVD-512	Random forest	{"max_features": 0.25, "min_samples_leaf": 1}	5
TNBC	SVD-512	XGBoost	{"learning_rate": 0.1, "max_depth": 3, "n_estimators": 300}	4
TNBC	SVD-512	XGBoost	{"learning_rate": 0.1, "max_depth": 6, "n_estimators": 300}	1

8 Temperature-scaling sensitivity

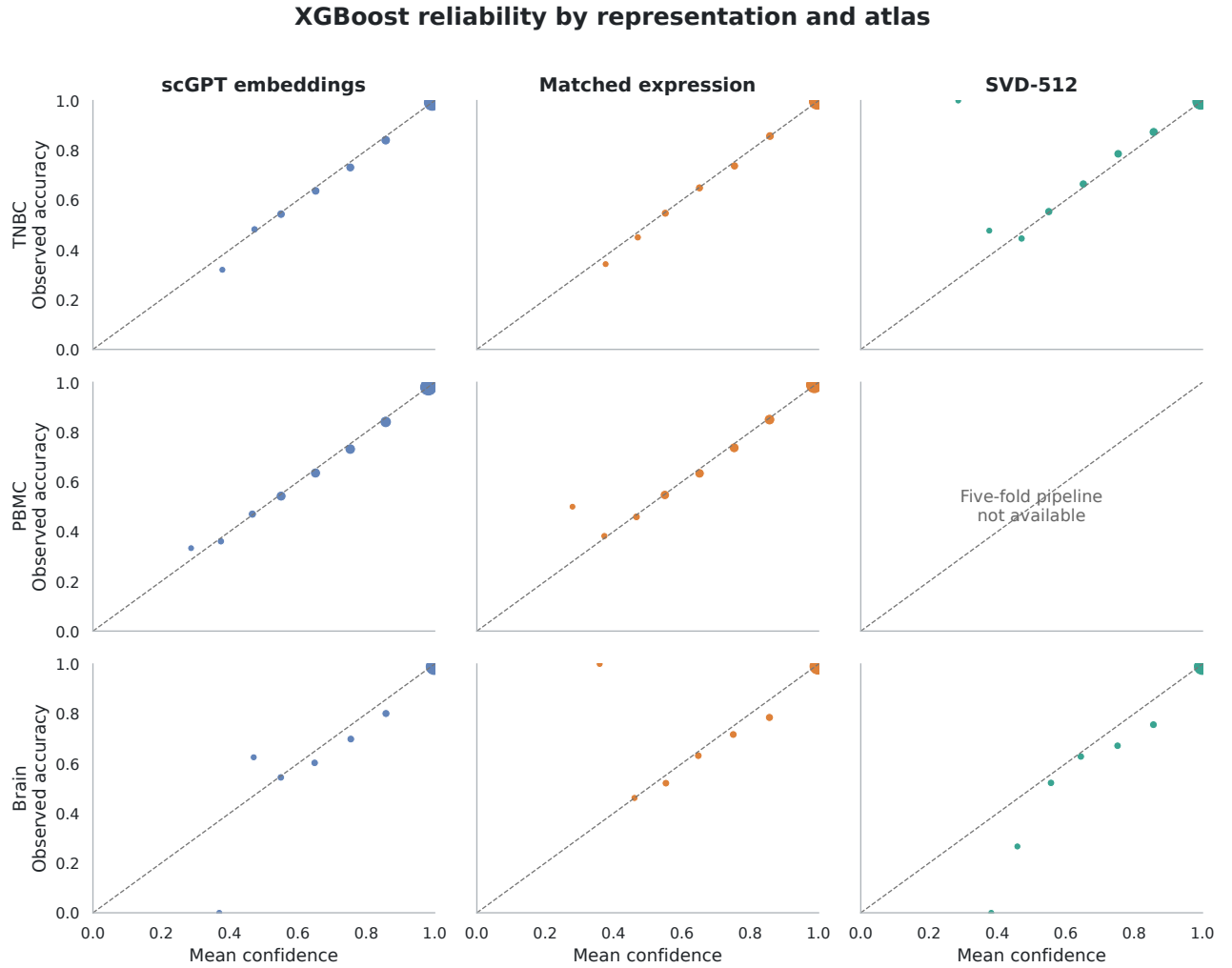
Table S8: Calibration metrics before and after scalar temperature scaling, summarized over outer folds.

Dataset	Representation	Classifier	State	T	ECE	Brier	NLL
Brain	Matched expression	Logistic regression	Before	1.329	0.011	0.026	0.087
Brain	Matched expression	Logistic regression	After	1.329	0.009	0.026	0.067
Brain	Matched expression	XGBoost	Before	1.250	0.011	0.026	0.083
Brain	Matched expression	XGBoost	After	1.250	0.011	0.026	0.072
Brain	scGPT embeddings	Logistic regression	Before	0.981	0.011	0.025	0.077
Brain	scGPT embeddings	Logistic regression	After	0.981	0.010	0.025	0.076
Brain	scGPT embeddings	XGBoost	Before	1.459	0.011	0.026	0.095
Brain	scGPT embeddings	XGBoost	After	1.459	0.009	0.026	0.073
Brain	SVD-512	Logistic regression	Before	1.337	0.011	0.027	0.093
Brain	SVD-512	Logistic regression	After	1.337	0.008	0.026	0.068
Brain	SVD-512	XGBoost	Before	1.487	0.011	0.025	0.094
Brain	SVD-512	XGBoost	After	1.487	0.010	0.025	0.071
TNBC	Matched expression	Logistic regression	Before	1.017	0.003	0.020	0.039
TNBC	Matched expression	Logistic regression	After	1.017	0.001	0.020	0.039
TNBC	Matched expression	XGBoost	Before	0.943	0.003	0.018	0.037
TNBC	Matched expression	XGBoost	After	0.943	0.002	0.018	0.037
TNBC	scGPT embeddings	Logistic regression	Before	1.035	0.003	0.033	0.066
TNBC	scGPT embeddings	Logistic regression	After	1.035	0.002	0.033	0.065
TNBC	scGPT embeddings	XGBoost	Before	1.065	0.006	0.041	0.080
TNBC	scGPT embeddings	XGBoost	After	1.065	0.003	0.041	0.078
TNBC	SVD-512	Logistic regression	Before	1.056	0.002	0.021	0.041
TNBC	SVD-512	Logistic regression	After	1.056	0.001	0.021	0.041
TNBC	SVD-512	XGBoost	Before	0.914	0.003	0.023	0.045
TNBC	SVD-512	XGBoost	After	0.914	0.001	0.023	0.045

9 Strict labelled-cell scarcity

Table S9: Strict Indonesia-PBMC label scarcity. Feature selection and tuning used only the labelled cells. Macro F1 is mean $\pm$ SD across five donor-held-out folds after averaging the two sampling seeds within fold.

Labels per class	scGPT embeddings	Matched expression	Mean matched HVGs
10	0.779 $\pm$ 0.032	0.818 $\pm$ 0.025	1169
50	0.841 $\pm$ 0.013	0.886 $\pm$ 0.019	1160
100	0.855 $\pm$ 0.009	0.904 $\pm$ 0.013	1156
250	0.872 $\pm$ 0.006	0.922 $\pm$ 0.006	1154
500	0.881 $\pm$ 0.008	0.930 $\pm$ 0.008	1149



Point area is proportional to the square root of the number of observations in the confidence bin; exact counts are supplied in CSV form.

Figure S2: Reliability of the final XGBoost pipelines across the human atlases. Point size encodes confidence-bin count; exact bin edges, counts, mean confidence and accuracy are supplied in `human_reliability_bins.csv`.

10 PBMC biological error profile

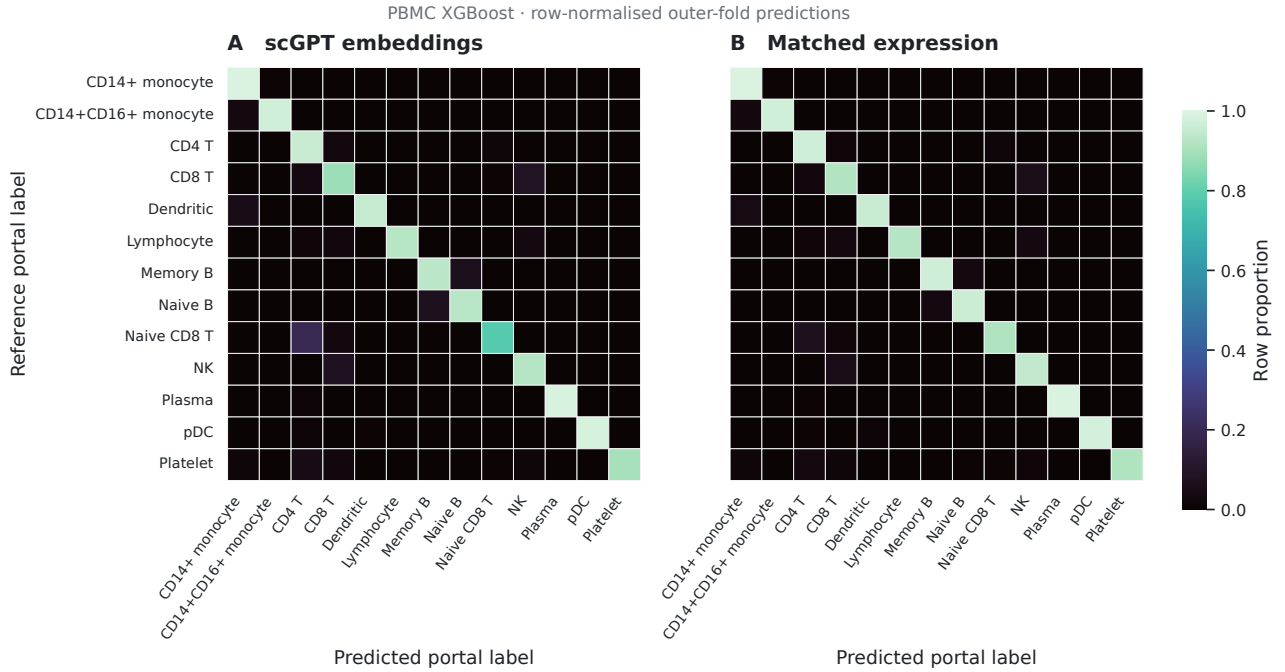


Figure S3: Row-normalized PBMC confusion matrices for XGBoost. Portal annotations are used as reference labels.

Table S10: Complete per-class metrics for PBMC.

Portal class	Representation	Classifier	Support	Precision	Recall	F1
CD4-positive, alpha-beta T cell	Matched expression	Logistic regression	135461	0.967	0.970	0.968
CD8-positive, alpha-beta T cell	Matched expression	Logistic regression	91629	0.922	0.920	0.921
natural killer cell	Matched expression	Logistic regression	81409	0.939	0.944	0.942
CD14-positive monocyte	Matched expression	Logistic regression	51039	0.990	0.992	0.991
memory B cell	Matched expression	Logistic regression	27172	0.968	0.968	0.968
naive thymus-derived	Matched expression	Logistic regression	26375	0.920	0.901	0.910
CD8-positive, alpha-beta T cell	Matched expression	Logistic regression	21672	0.961	0.961	0.961
naive B cell						
CD14-positive, CD16-positive monocyte						
dendritic cell, human	Matched expression	Logistic regression	4374	0.967	0.965	0.966
platelet	Matched expression	Logistic regression	3013	0.929	0.924	0.926
plasma cell	Matched expression	Logistic regression	2425	0.993	0.989	0.991
plasmacytoid dendritic cell	Matched expression	Logistic regression	2005	0.994	0.986	0.990
lymphocyte	Matched expression	Logistic regression	1957	0.967	0.943	0.955
CD4-positive, alpha-beta T cell	scGPT embeddings	Logistic regression	135461	0.954	0.961	0.957
CD8-positive, alpha-beta T cell	scGPT embeddings	Logistic regression	91629	0.906	0.902	0.904
natural killer cell	scGPT embeddings	Logistic regression	81409	0.925	0.934	0.930
CD14-positive monocyte	scGPT embeddings	Logistic regression	51039	0.988	0.990	0.989
memory B cell	scGPT embeddings	Logistic regression	27172	0.955	0.951	0.953
naive thymus-derived	scGPT embeddings	Logistic regression	26375	0.900	0.847	0.873
CD8-positive, alpha-beta T cell	scGPT embeddings	Logistic regression	21672	0.939	0.945	0.942
naive B cell						
CD14-positive, CD16-positive monocyte						
dendritic cell, human	scGPT embeddings	Logistic regression	4374	0.962	0.959	0.961
platelet	scGPT embeddings	Logistic regression	3013	0.927	0.923	0.925
plasma cell	scGPT embeddings	Logistic regression	2425	0.994	0.993	0.994
plasmacytoid dendritic cell	scGPT embeddings	Logistic regression	2005	0.989	0.988	0.988
lymphocyte	scGPT embeddings	Logistic regression	1957	0.957	0.947	0.952
CD4-positive, alpha-beta T cell	SVD-512	Logistic regression	135461	0.966	0.970	0.968
CD8-positive, alpha-beta T cell	SVD-512	Logistic regression	91629	0.919	0.916	0.918
natural killer cell	SVD-512	Logistic regression	81409	0.935	0.940	0.937
CD14-positive monocyte	SVD-512	Logistic regression	51039	0.990	0.991	0.990
memory B cell	SVD-512	Logistic regression	27172	0.968	0.968	0.968

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Table S10 continued

Portal class	Representation	Classifier	Support	Precision	Recall	F1
naive thymus-derived	SVD-512	Logistic regression	26375	0.920	0.901	0.910
CD8-positive, alpha-beta T cell						
naive B cell	SVD-512	Logistic regression	21672	0.960	0.961	0.960
CD14-positive, CD16-positive	SVD-512	Logistic regression	13503	0.981	0.978	0.979
monocyte						
dendritic cell, human	SVD-512	Logistic regression	4374	0.962	0.961	0.962
platelet	SVD-512	Logistic regression	3013	0.930	0.919	0.924
plasma cell	SVD-512	Logistic regression	2425	0.993	0.993	0.993
plasmacytoid dendritic cell	SVD-512	Logistic regression	2005	0.991	0.986	0.989
lymphocyte	SVD-512	Logistic regression	1957	0.964	0.934	0.948
CD4-positive, alpha-beta T cell	Matched expression	Random forest	135461	0.964	0.963	0.964
CD8-positive, alpha-beta T cell	Matched expression	Random forest	91629	0.902	0.904	0.903
natural killer cell	Matched expression	Random forest	81409	0.919	0.933	0.926
CD14-positive monocyte	Matched expression	Random forest	51039	0.981	0.990	0.986
memory B cell	Matched expression	Random forest	27172	0.953	0.949	0.951
naive thymus-derived	Matched expression	Random forest	26375	0.903	0.902	0.903
CD8-positive, alpha-beta T cell						
naive B cell	Matched expression	Random forest	21672	0.941	0.938	0.940
CD14-positive, CD16-positive	Matched expression	Random forest	13503	0.977	0.955	0.966
monocyte						
dendritic cell, human	Matched expression	Random forest	4374	0.968	0.935	0.951
platelet	Matched expression	Random forest	3013	0.949	0.740	0.831
plasma cell	Matched expression	Random forest	2425	0.997	0.979	0.988
plasmacytoid dendritic cell	Matched expression	Random forest	2005	0.996	0.947	0.971
lymphocyte	Matched expression	Random forest	1957	0.991	0.749	0.853
CD4-positive, alpha-beta T cell	scGPT embeddings	Random forest	135461	0.897	0.957	0.926
CD8-positive, alpha-beta T cell	scGPT embeddings	Random forest	91629	0.868	0.857	0.862
natural killer cell	scGPT embeddings	Random forest	81409	0.899	0.911	0.905
CD14-positive monocyte	scGPT embeddings	Random forest	51039	0.982	0.990	0.986
memory B cell	scGPT embeddings	Random forest	27172	0.923	0.919	0.921
naive thymus-derived	scGPT embeddings	Random forest	26375	0.901	0.616	0.732
CD8-positive, alpha-beta T cell						
naive B cell	scGPT embeddings	Random forest	21672	0.901	0.905	0.903
CD14-positive, CD16-positive	scGPT embeddings	Random forest	13503	0.970	0.964	0.967
monocyte						
dendritic cell, human	scGPT embeddings	Random forest	4374	0.968	0.938	0.953
platelet	scGPT embeddings	Random forest	3013	0.928	0.829	0.875
plasma cell	scGPT embeddings	Random forest	2425	0.997	0.980	0.988
plasmacytoid dendritic cell	scGPT embeddings	Random forest	2005	0.993	0.974	0.983
lymphocyte	scGPT embeddings	Random forest	1957	0.978	0.823	0.893
CD4-positive, alpha-beta T cell	Matched expression	XGBoost	135461	0.969	0.968	0.968
CD8-positive, alpha-beta T cell	Matched expression	XGBoost	91629	0.922	0.918	0.920
natural killer cell	Matched expression	XGBoost	81409	0.936	0.945	0.940
CD14-positive monocyte	Matched expression	XGBoost	51039	0.988	0.992	0.990
memory B cell	Matched expression	XGBoost	27172	0.970	0.967	0.969
naive thymus-derived	Matched expression	XGBoost	26375	0.915	0.912	0.913
CD8-positive, alpha-beta T cell						
naive B cell	Matched expression	XGBoost	21672	0.961	0.961	0.961
CD14-positive, CD16-positive	Matched expression	XGBoost	13503	0.981	0.972	0.976
monocyte						
dendritic cell, human	Matched expression	XGBoost	4374	0.968	0.955	0.961
platelet	Matched expression	XGBoost	3013	0.924	0.914	0.919
plasma cell	Matched expression	XGBoost	2425	0.995	0.989	0.992
plasmacytoid dendritic cell	Matched expression	XGBoost	2005	0.994	0.977	0.985
lymphocyte	Matched expression	XGBoost	1957	0.973	0.921	0.946
CD4-positive, alpha-beta T cell	scGPT embeddings	XGBoost	135461	0.936	0.955	0.945
CD8-positive, alpha-beta T cell	scGPT embeddings	XGBoost	91629	0.884	0.880	0.882
natural killer cell	scGPT embeddings	XGBoost	81409	0.909	0.921	0.915
CD14-positive monocyte	scGPT embeddings	XGBoost	51039	0.986	0.989	0.987
memory B cell	scGPT embeddings	XGBoost	27172	0.941	0.933	0.937
naive thymus-derived	scGPT embeddings	XGBoost	26375	0.891	0.778	0.830
CD8-positive, alpha-beta T cell						
naive B cell	scGPT embeddings	XGBoost	21672	0.918	0.927	0.923
CD14-positive, CD16-positive	scGPT embeddings	XGBoost	13503	0.970	0.969	0.970
monocyte						
dendritic cell, human	scGPT embeddings	XGBoost	4374	0.963	0.946	0.954
platelet	scGPT embeddings	XGBoost	3013	0.909	0.895	0.902
plasma cell	scGPT embeddings	XGBoost	2425	0.991	0.985	0.988
plasmacytoid dendritic cell	scGPT embeddings	XGBoost	2005	0.990	0.980	0.985
lymphocyte	scGPT embeddings	XGBoost	1957	0.958	0.922	0.940

11 Brain biological error profile

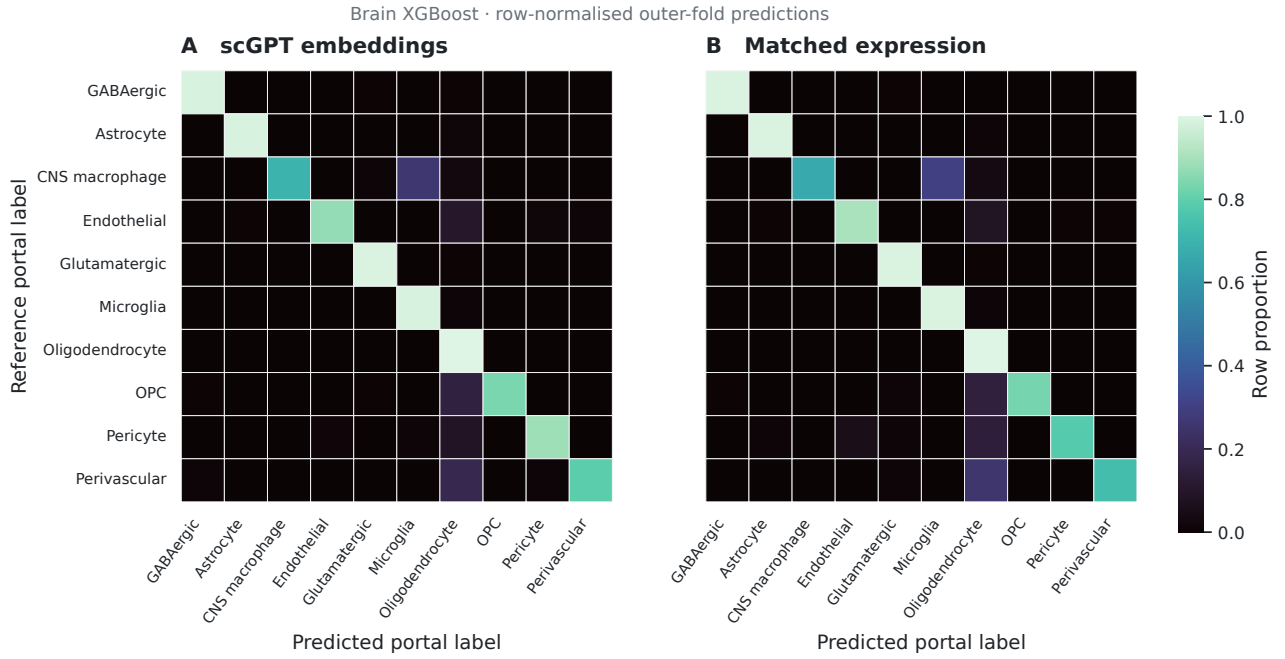


Figure S4: Row-normalized Brain confusion matrices for XGBoost. Portal annotations are used as reference labels.

Table S11: Complete per-class metrics for Brain.

Portal class	Representation	Classifier	Support	Precision	Recall	F1
oligodendrocyte	Matched expression	Logistic regression	37430	0.981	0.995	0.988
glutamatergic neuron	Matched expression	Logistic regression	18177	0.994	0.991	0.993
GABAergic neuron	Matched expression	Logistic regression	8310	0.989	0.990	0.989
astrocyte	Matched expression	Logistic regression	5042	0.995	0.990	0.993
microglial cell	Matched expression	Logistic regression	3038	0.982	0.984	0.983
oligodendrocyte precursor cell	Matched expression	Logistic regression	3014	0.950	0.833	0.888
endothelial cell	Matched expression	Logistic regression	247	0.968	0.850	0.905
central nervous system	Matched expression	Logistic regression	113	0.769	0.619	0.686
macrophage						
perivascular cell	Matched expression	Logistic regression	105	0.987	0.714	0.829
pericyte	Matched expression	Logistic regression	104	0.976	0.769	0.860
oligodendrocyte	scGPT embeddings	Logistic regression	37430	0.983	0.997	0.990
glutamatergic neuron	scGPT embeddings	Logistic regression	18177	0.994	0.992	0.993
GABAergic neuron	scGPT embeddings	Logistic regression	8310	0.988	0.990	0.989
astrocyte	scGPT embeddings	Logistic regression	5042	0.993	0.990	0.991
microglial cell	scGPT embeddings	Logistic regression	3038	0.981	0.992	0.987
oligodendrocyte precursor cell	scGPT embeddings	Logistic regression	3014	0.976	0.829	0.897
endothelial cell	scGPT embeddings	Logistic regression	247	0.970	0.911	0.939
central nervous system	scGPT embeddings	Logistic regression	113	0.940	0.690	0.796
macrophage						
perivascular cell	scGPT embeddings	Logistic regression	105	0.947	0.848	0.894
pericyte	scGPT embeddings	Logistic regression	104	0.931	0.913	0.922
oligodendrocyte	SVD-512	Logistic regression	37430	0.981	0.996	0.988
glutamatergic neuron	SVD-512	Logistic regression	18177	0.994	0.991	0.993
GABAergic neuron	SVD-512	Logistic regression	8310	0.989	0.989	0.989
astrocyte	SVD-512	Logistic regression	5042	0.995	0.988	0.992
microglial cell	SVD-512	Logistic regression	3038	0.981	0.980	0.981
oligodendrocyte precursor cell	SVD-512	Logistic regression	3014	0.954	0.833	0.890
endothelial cell	SVD-512	Logistic regression	247	0.963	0.834	0.894
central nervous system	SVD-512	Logistic regression	113	0.704	0.611	0.654
macrophage						
perivascular cell	SVD-512	Logistic regression	105	0.986	0.695	0.816
pericyte	SVD-512	Logistic regression	104	0.954	0.798	0.869
oligodendrocyte	Matched expression	Random forest	37430	0.975	0.998	0.987
glutamatergic neuron	Matched expression	Random forest	18177	0.997	0.984	0.991
GABAergic neuron	Matched expression	Random forest	8310	0.986	0.990	0.988

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Table S11 continued

Portal class	Representation	Classifier	Support	Precision	Recall	F1
astrocyte	Matched expression	Random forest	5042	0.995	0.986	0.991
microglial cell	Matched expression	Random forest	3038	0.974	0.990	0.982
oligodendrocyte precursor cell	Matched expression	Random forest	3014	0.991	0.828	0.902
endothelial cell	Matched expression	Random forest	247	0.963	0.729	0.829
central nervous system	Matched expression	Random forest	113	0.958	0.407	0.571
macrophage						
perivascular cell	Matched expression	Random forest	105	0.982	0.533	0.691
pericyte	Matched expression	Random forest	104	0.985	0.625	0.765
oligodendrocyte	scGPT embeddings	Random forest	37430	0.977	0.998	0.987
glutamatergic neuron	scGPT embeddings	Random forest	18177	0.994	0.989	0.991
GABAergic neuron	scGPT embeddings	Random forest	8310	0.991	0.980	0.985
astrocyte	scGPT embeddings	Random forest	5042	0.994	0.986	0.990
microglial cell	scGPT embeddings	Random forest	3038	0.982	0.989	0.986
oligodendrocyte precursor cell	scGPT embeddings	Random forest	3014	0.992	0.830	0.904
endothelial cell	scGPT embeddings	Random forest	247	0.976	0.810	0.885
central nervous system	scGPT embeddings	Random forest	113	0.914	0.655	0.763
macrophage						
perivascular cell	scGPT embeddings	Random forest	105	0.971	0.648	0.777
pericyte	scGPT embeddings	Random forest	104	0.973	0.702	0.816
oligodendrocyte	SVD-512	Random forest	37430	0.980	0.998	0.989
glutamatergic neuron	SVD-512	Random forest	18177	0.995	0.991	0.993
GABAergic neuron	SVD-512	Random forest	8310	0.988	0.992	0.990
astrocyte	SVD-512	Random forest	5042	0.993	0.994	0.993
microglial cell	SVD-512	Random forest	3038	0.964	0.990	0.977
oligodendrocyte precursor cell	SVD-512	Random forest	3014	0.998	0.835	0.909
endothelial cell	SVD-512	Random forest	247	0.969	0.623	0.759
central nervous system	SVD-512	Random forest	113	0.920	0.204	0.333
macrophage						
perivascular cell	SVD-512	Random forest	105	0.971	0.629	0.763
pericyte	SVD-512	Random forest	104	0.971	0.644	0.775
oligodendrocyte	Matched expression	XGBoost	37430	0.981	0.998	0.990
glutamatergic neuron	Matched expression	XGBoost	18177	0.996	0.991	0.994
GABAergic neuron	Matched expression	XGBoost	8310	0.989	0.992	0.990
astrocyte	Matched expression	XGBoost	5042	0.994	0.991	0.992
microglial cell	Matched expression	XGBoost	3038	0.982	0.988	0.985
oligodendrocyte precursor cell	Matched expression	XGBoost	3014	0.984	0.831	0.901
endothelial cell	Matched expression	XGBoost	247	0.970	0.903	0.935
central nervous system	Matched expression	XGBoost	113	0.938	0.664	0.777
macrophage						
perivascular cell	Matched expression	XGBoost	105	0.975	0.733	0.837
pericyte	Matched expression	XGBoost	104	0.942	0.779	0.853
oligodendrocyte	scGPT embeddings	XGBoost	37430	0.981	0.997	0.989
glutamatergic neuron	scGPT embeddings	XGBoost	18177	0.995	0.991	0.993
GABAergic neuron	scGPT embeddings	XGBoost	8310	0.990	0.988	0.989
astrocyte	scGPT embeddings	XGBoost	5042	0.994	0.988	0.991
microglial cell	scGPT embeddings	XGBoost	3038	0.982	0.988	0.985
oligodendrocyte precursor cell	scGPT embeddings	XGBoost	3014	0.984	0.833	0.902
endothelial cell	scGPT embeddings	XGBoost	247	0.968	0.870	0.917
central nervous system	scGPT embeddings	XGBoost	113	0.868	0.699	0.775
macrophage						
perivascular cell	scGPT embeddings	XGBoost	105	0.922	0.790	0.851
pericyte	scGPT embeddings	XGBoost	104	0.939	0.885	0.911
oligodendrocyte	SVD-512	XGBoost	37430	0.983	0.998	0.990
glutamatergic neuron	SVD-512	XGBoost	18177	0.995	0.993	0.994
GABAergic neuron	SVD-512	XGBoost	8310	0.989	0.992	0.990
astrocyte	SVD-512	XGBoost	5042	0.995	0.992	0.993
microglial cell	SVD-512	XGBoost	3038	0.975	0.994	0.984
oligodendrocyte precursor cell	SVD-512	XGBoost	3014	0.986	0.836	0.905
endothelial cell	SVD-512	XGBoost	247	0.970	0.789	0.871
central nervous system	SVD-512	XGBoost	113	0.921	0.513	0.659
macrophage						
perivascular cell	SVD-512	XGBoost	105	0.945	0.657	0.775
pericyte	SVD-512	XGBoost	104	0.976	0.798	0.878

12 Pig-pancreas cross-species stress test

The porcine object contained one donor ID; this analysis therefore uses a random cell split and is excluded from the human primary summary. The whole-human checkpoint was applied by exact/uppercase gene-symbol matching without an orthology map. Results should be read only as a stress test.

Table S12: Pig-pancreas stress-test metrics.

Representation	Classifier	Train	Test	Macro F1	Accuracy	ECE	Brier	NLL
Matched expression	Logistic regression	17644	4412	0.988	0.997	0.002	0.005	0.012
scGPT embeddings	Logistic regression	17644	4412	0.966	0.993	0.002	0.011	0.024
SVD-512	Logistic regression	17644	4412	0.987	0.997	0.003	0.004	0.008
Matched expression	Random forest	17644	4412	0.989	0.995	0.005	0.008	0.017
scGPT embeddings	Random forest	17644	4412	0.613	0.943	0.049	0.090	0.183
SVD-512	Random forest	17644	4412	0.938	0.995	0.011	0.010	0.025
Matched expression	XGBoost	17644	4412	0.992	0.997	0.002	0.006	0.011
scGPT embeddings	XGBoost	17644	4412	0.888	0.974	0.011	0.040	0.084
SVD-512	XGBoost	17644	4412	0.967	0.996	0.002	0.006	0.012

13 Machine-readable audit material

The reproducibility archive contains the complete confusion matrices, reliability-bin counts, strict-scarcity seed-level results, selected genes for every outer fold, SVD explained-variance audits, retained hyperparameter traces, model timings, software versions, checkpoint hashes and the original-index alignment records used to verify the out-of-fold partition. The pipeline-coverage table identifies every representation–classifier combination admitted to aggregation.